

Unit 11, Lesson 5: Writing Exponential Functions From a Graph or Table



Benchmark

MA.912.AR.5.4 Write an exponential function to represent a relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context.



Learning Targets

- ☐ I can write an exponential function from a graph.
- ☐ I can write an exponential function from a table.



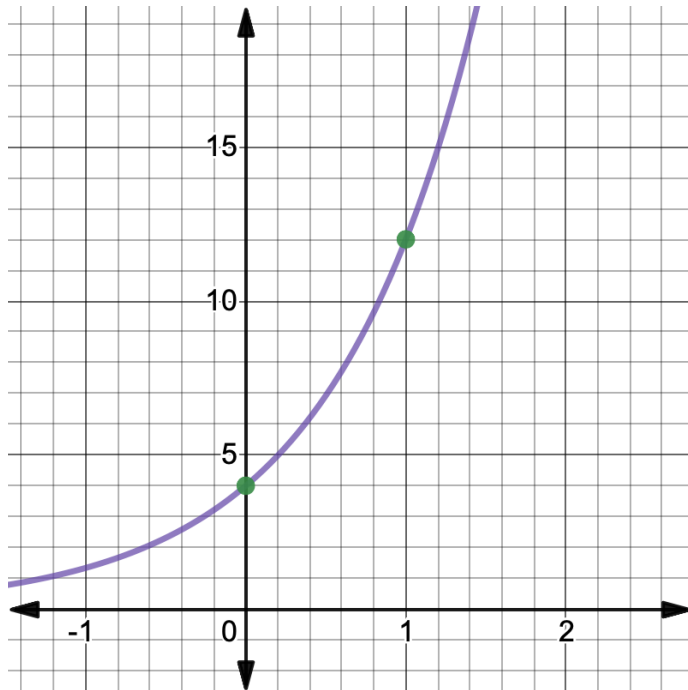
11.5.1 Warm-Up: Growth Mindset

1. It's crucial to create an environment where failure, errors, and mistakes are seen as a natural part of learning. Making that environment all starts with you; more specifically, your self-talk.
 - a. When you make a mistake, what do you say to yourself?
 - b. Explain if what you say to yourself reflects a growth or fixed mindset.



11.5.2 Guided Instruction: Writing an Exponential Function From Graphs and Tables

1. The graph of an exponential function is shown.



To express this exponential function in the form $f(x) = a(b)^x$ or $f(x) = a(1 \pm r)^x$, the values for a and b need to be found. These values can be found using two points on the graph and the equation in the form $y = a(b)^x$.

- a. Find the value of a by substituting point $(0, 4)$ in for x and y in $y = a(b)^x$.

$$\boxed{} = a(b)^{\boxed{}}$$

- b. Discuss with your partner if the point $(1, 12)$ can be used to determine the value of b . Write a brief summary of your discussion.

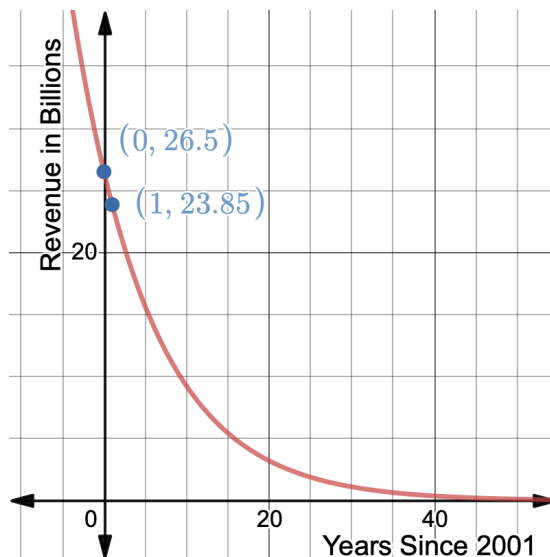
- c. Use the value of a found in Part A and the point $(1, 12)$ to determine the value of b .

$$\boxed{} = \boxed{}(b)^{\boxed{}}$$

- d. Write the function for the graph using both forms.

$f(x) = a(b)^x$	$f(x) = a(1 \pm r)^x$

2. The Global Music Report shows there has been a decline in revenue from physical records due to the increase of digital downloads and streaming services. The graph shows y , the revenue in billions, and x , years since 2001.



- a. Discuss with your partner which point can be used to find the a value for the equation $y = a(b)^x$.
- b. Work with your partner to write the equation for the revenue of physical records since 2001.

c. What does the a value represent in the context?

d. What does the b value represent in the context?

3. Writing an equation from a table of values can be done by finding the initial value, a , and the common ratio, b .

The table of values represents an exponential relationship.

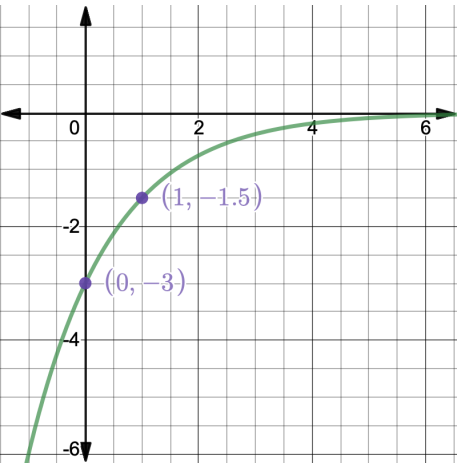
x	0	1	2	3	4	5	6
$f(x)$	$\frac{2}{3}$	2	6	18	54	162	486



- a. Describe how the values of $f(x)$ change as the x -values increase by 1.
- b. What is the initial value, a ?

- c. Discuss with your partner how to use the information determined in Parts A and B to write the function in the form $f(x) = a(b)^x$. Then, write the function.

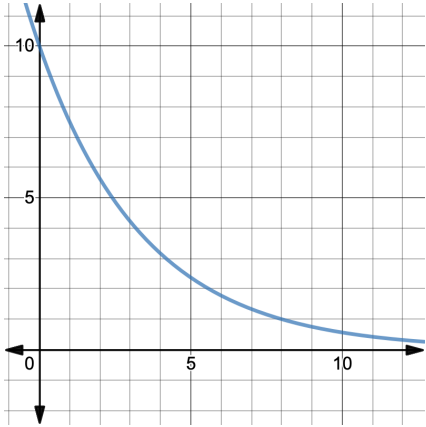
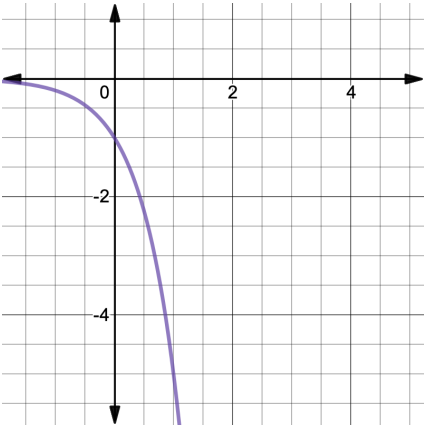
4. When the value of a is negative, the same process can be used. Write an exponential function for the graph and for the table of values in the form $f(x) = a(b)^x$.

Graph	Table												
	<table border="1"> <thead> <tr> <th>x</th><th>$f(x)$</th></tr> </thead> <tbody> <tr> <td>0</td><td>-2</td></tr> <tr> <td>1</td><td>-7</td></tr> <tr> <td>2</td><td>-24.5</td></tr> <tr> <td>3</td><td>-85.75</td></tr> <tr> <td>4</td><td>-300.125</td></tr> </tbody> </table>	x	$f(x)$	0	-2	1	-7	2	-24.5	3	-85.75	4	-300.125
x	$f(x)$												
0	-2												
1	-7												
2	-24.5												
3	-85.75												
4	-300.125												
Function	Function												



11.5.3 Guided Practice: Writing Exponential Functions

1. Write the exponential function for each graph.

	
Function	Function

2. Write an exponential function for each table of values.

x	$f(x)$
0	4
1	10
2	25
3	62.5
4	156.25

x	$f(x)$
0	20
1	16
2	$\frac{64}{5}$
3	$\frac{256}{25}$
4	$\frac{1,024}{125}$

Function

Function



11.5.4 Wrap-Up: Ticket Out the Door

1. Write the exponential function represented by the graph and the exponential function represented in the table of values.

Graph	Table												
	<table> <tr> <th>x</th><th>$f(x)$</th></tr> <tr> <td>0</td><td>0.7</td></tr> <tr> <td>1</td><td>3.5</td></tr> <tr> <td>2</td><td>17.5</td></tr> <tr> <td>3</td><td>87.5</td></tr> <tr> <td>4</td><td>437.5</td></tr> </table>	x	$f(x)$	0	0.7	1	3.5	2	17.5	3	87.5	4	437.5
x	$f(x)$												
0	0.7												
1	3.5												
2	17.5												
3	87.5												
4	437.5												
Function	Function												
$f(x) =$	$f(x) =$												